

Algorithms Problem Solving

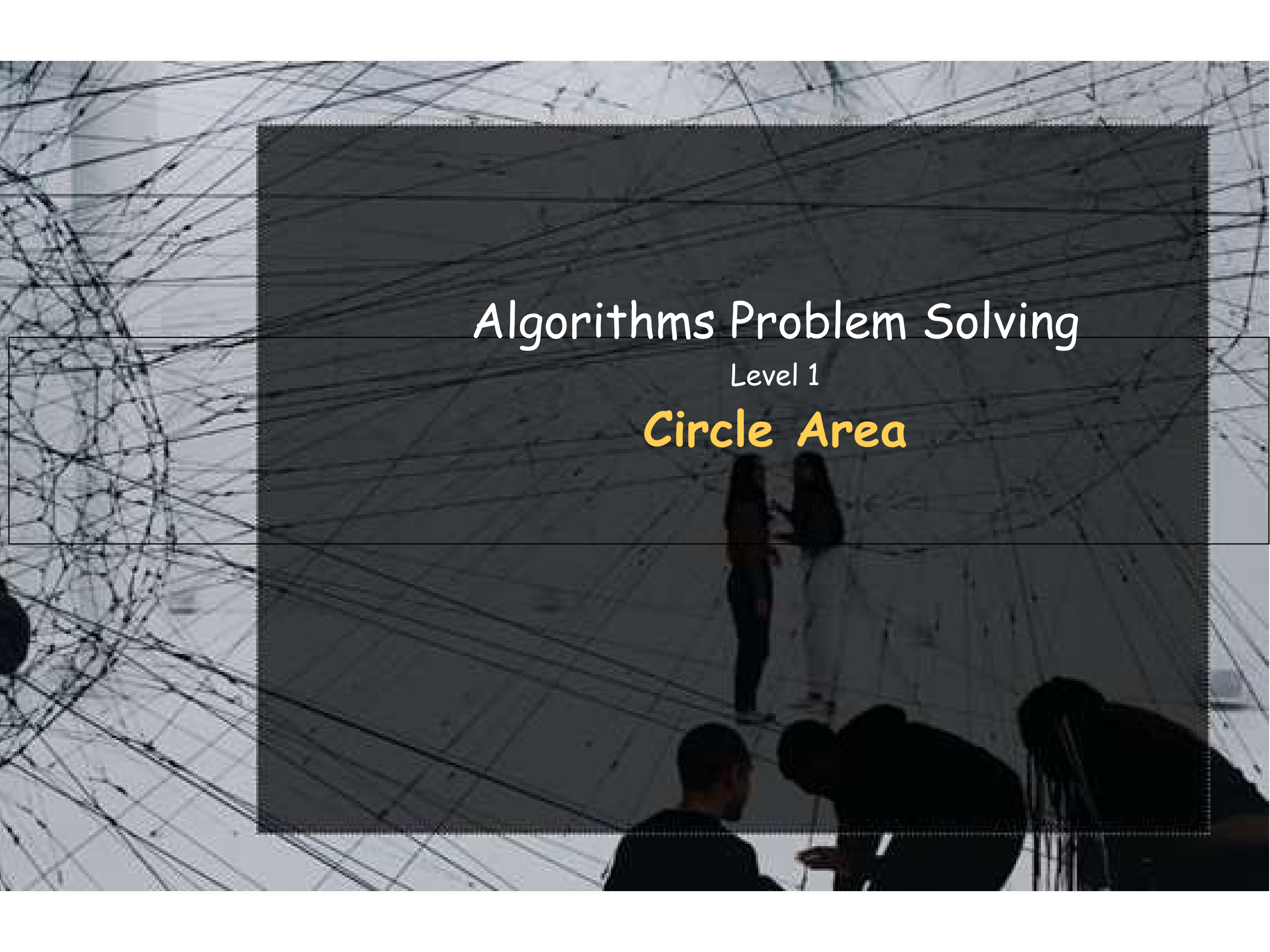
Level 1

Circle Area

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Definition

- ◆ This program reads the radius of a circle and calculates its area.
- ◆ The program uses functions, a mathematical constant, `pow()`, multiplication, and float values.

Explanation

- ◆ The `readNumbers()` function reads the circle radius `R` from the user.
- ◆ The `CircleArea()` function calculates the area using the circle area formula:
- ◆ $\text{Area} = \pi \times R^2$
- ◆ `M_PI` represents the mathematical constant π .
- ◆ `pow(R, 2)` calculates the radius squared.
- ◆ The result is stored in a float variable called `Area` and then returned.
- ◆ The `printTriangleArea()` function calls `CircleArea()` and prints the calculated area.
- ◆ Although the function is named `printTriangleArea()`, it actually prints the circle area. A clearer name would be `printCircleArea()`.

Steps in main()

1. Read the circle radius.
2. Calculate the radius squared using `pow()`.
3. Multiply the result by π .
4. Store the calculated area.
5. Return the area.
6. Print the circle area.

💡 Example

◆ If the user enters:

Radius: 5

◆ Calculate the area:

$$\text{Area} = \pi \times 5^2$$

$$\text{Area} = \pi \times 25$$

◆ The result is approximately 78.5398.

➡ Result

Circle Area = 78.5398

Note

- ◆ M_PI is a mathematical constant representing π .
- ◆ The program includes `<cmath>` for mathematical functions such as `pow()`.
- ◆ The header `<corecrt_math_defines.h>` is used here to provide M_PI in this environment.

⊘ Warning

◆ The function name `printTriangleArea()` does not match what the function actually does. It would be clearer to rename it to `printCircleArea()`.



Comparison

▨ pow(R, 2)

- ✓ Calculates R^2 .
- ✓ Used as part of the circle area formula.
- ✓ Returns the squared value.

▤ M_PI

- ✓ Represents π .
- ✓ Is a mathematical constant.
- ✓ Is multiplied by R^2 to calculate the area.

✦ Summary

- ✓ The area of a circle is calculated using $\pi \times R^2$.
- ✓ `pow(R, 2)` calculates the radius squared.
- ✓ `M_PI` represents the mathematical constant π .
- ✓ The calculation returns a float value because the result can contain decimals.
- ✓ Function names should describe what the function actually does.

Thank you 😊